

ACHINTYA RAI

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EDUCATION

University of California, San Diego

Bachelor of Science in Computer Engineering

Sept. 2024 – June 2027

GPA: 3.92/4.0

- Competitive Programming Club, NSF REU Fellow (National Science Foundation), UC Scholars Research Fellow

WORK EXPERIENCE

Adobe

Incoming Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered Flask backend and database schemas to unify **100k+** datapoints across 20+ distinct fitness sources.
- Implemented asynchronous batch processing and storage workflows in Google Cloud Platform, yielding a **35%** increase in execution speed and **60%** reduction in redundant database write operations.
- Automated environment setup via Bash scripts, reducing developer testing time by **80%**.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Optimized Python/NumPy pipelines for Amazon/Coca-Cola, boosting capacity **15%** for **1M+** weekly events.
- Engineered visualization data structures, resulting in **20%** faster reporting and improved dashboard data integrity.
- Developed automated data-validation scripts to catch ingestion errors, reducing manual debugging time by **25%**.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

Mar. 2026 – Present

- Engineered a Python pipeline reducing LLM manifold footprints from **500GB** to **13GB** (Llama-2/DeepSeek).
- Optimized architectural placement scripts, achieving a **30x runtime reduction** (30m to <1m) on resource-constrained **8GB RAM** systems.
- Identified model "backbones" via Louvain Clustering, increasing ICN density from **0.18%** to **13.25%** in Llama-2-70B through noise-filtering.

Data Science Teaching & Learning Lab

Undergraduate Researcher

San Diego, CA

Sept. 2024 – Present

- Developed an LLM-augmented data tool using FastAPI and Gemini API to automate labeling for **16k+** data points, reducing manual instructor workload by **90%**.
- Architected a multi-agent framework, implementing hallucination-reduction strategies to achieve **95%+ accuracy** in real-time academic response generation.
- Engineered agentic workflows to synthesize over **100k+** data points from Jupyter Notebooks for real-time insights.

PROJECTS

Watchdog | Python, Gemini Vision, Browser-Use, Asyncio, SQLite

April 2026

- Engineered an agentic safeguard to autonomously monitor and manage browser tabs via GUI automation.
- Built a concurrent CLI for real-time **Notion API** sync and **aiosqlite** non-blocking state persistence.

SquaredUpChess | React, Django, AWS, Docker, WebSockets, Coolify, SQLite

Aug. 2025 – Dec. 2025

- Engineered a full-stack platform on **AWS** for real-time coach-student scheduling and secure payment processing.
- Architected **20+ Django REST endpoints** with a **CI/CD pipeline** to ensure 99.9% deployment reliability.

NanoGPT | Python, PyTorch, Transformers, LLM Optimization

Mar. 2026

- Implemented a GPT-style transformer from scratch with **multi-headed attention** and **backpropagation** logic.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, SQL (Postgres/SQLite), Java, JavaScript, Bash, HTML/CSS

Development: React, Django, Node.js, FastAPI, Asyncio, WebSockets, Flask, JUnit, PyTest

AI & Cloud: PyTorch, AWS, GCP, Gemini/OpenAI API, Docker, GitHub Actions, browser-use, Linux